Colour perception

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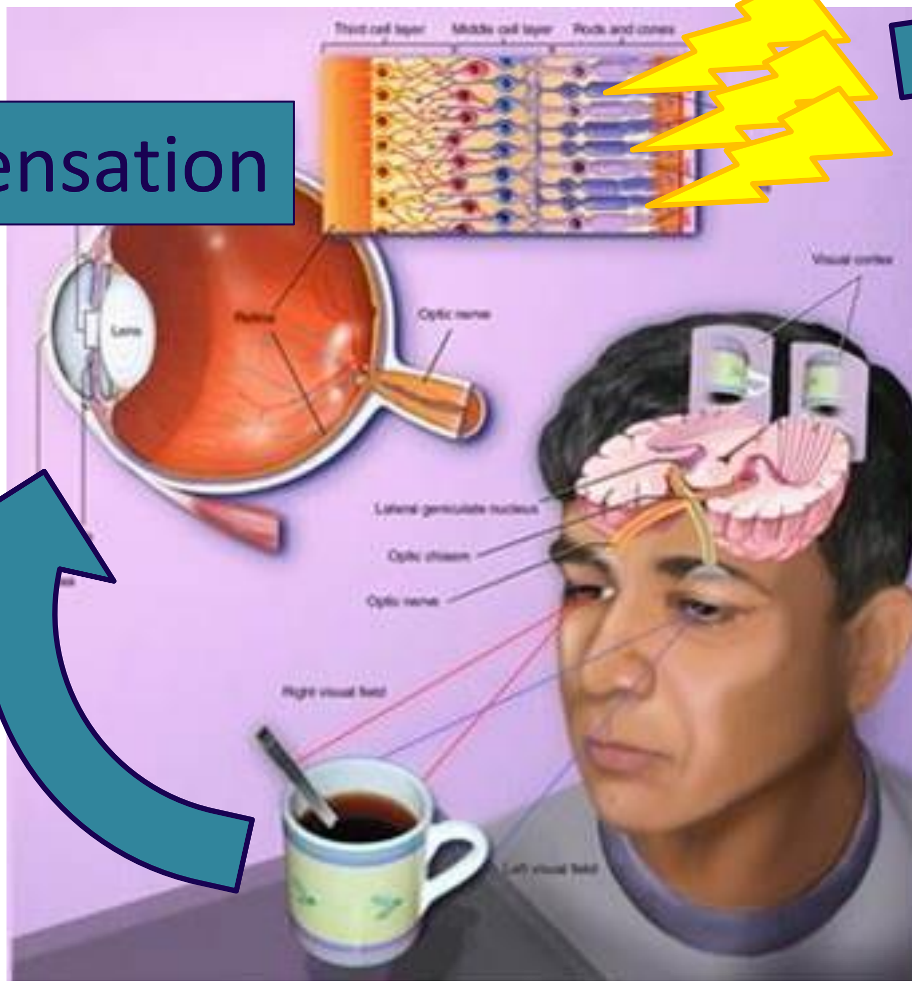
Last 2 weeks…

Retinal images & electrical signals created by the detection of a stimulus

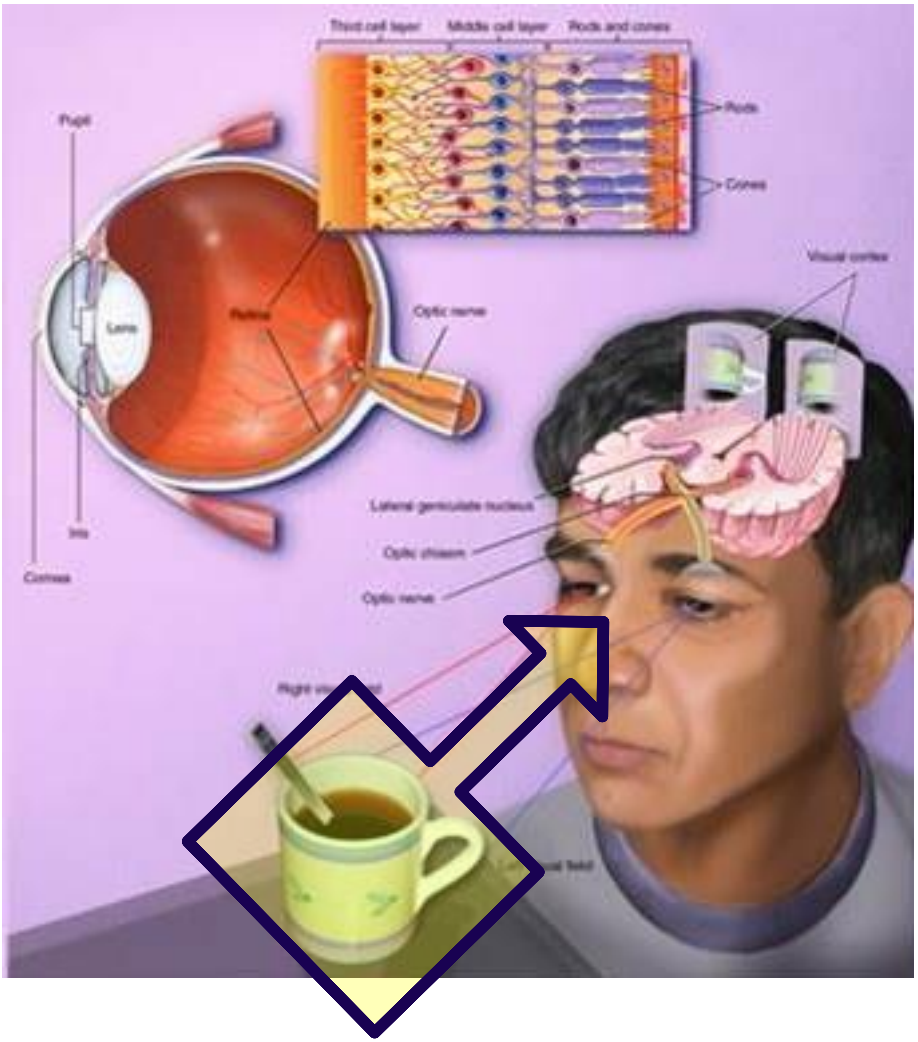
Sensation



Perception

Making sense of retinal images and electrical signals

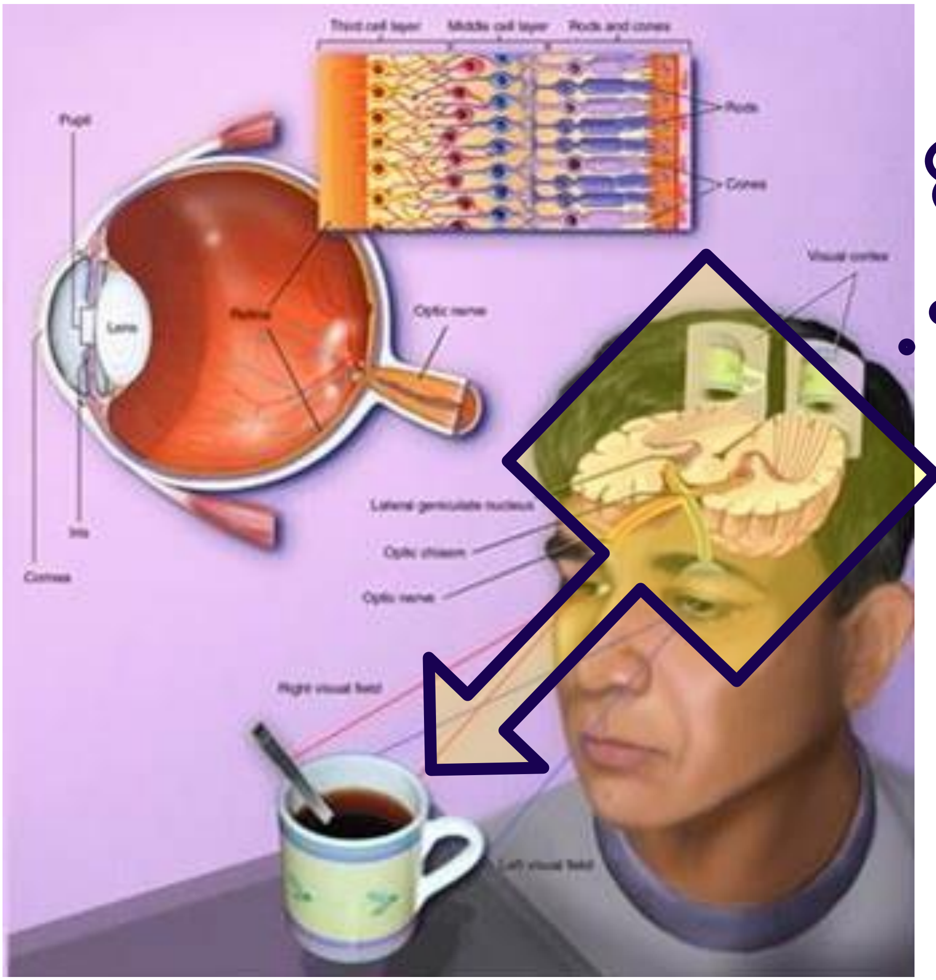
Last week…



Perception

Bottom-up guided perception
The sensory input determines what is perceived:
- describing (*structuralism*) and measuring (*psychophysics*) the elements of perception
- affordance of objects (*Gibson*)

Last week…



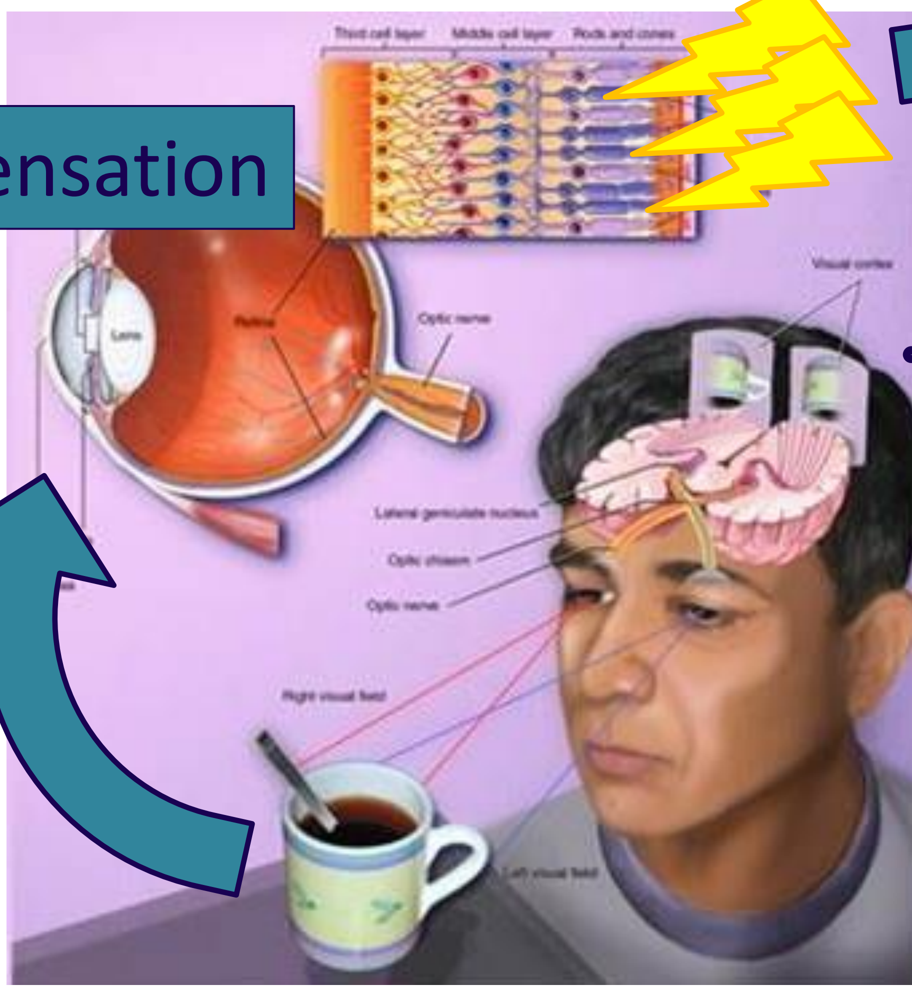
Cognition

Perception

Top-down guided perception
The sensory input is interpreted to be perceived:
- constructing perception by testing hypotheses (*Gregory*)
- grouping and segmenting the sensory input with perceptual heuristics (*Gestalt Psychology*)

This and next week…

Sensation



Cognition

Perception

- Colour
- Depth & size
- Motion

Today's topic!

What does colour do for us?

- Colour enhances
  - Perceptual organisation.

  - Top-down control.
  - Automatic bottom-up processing.

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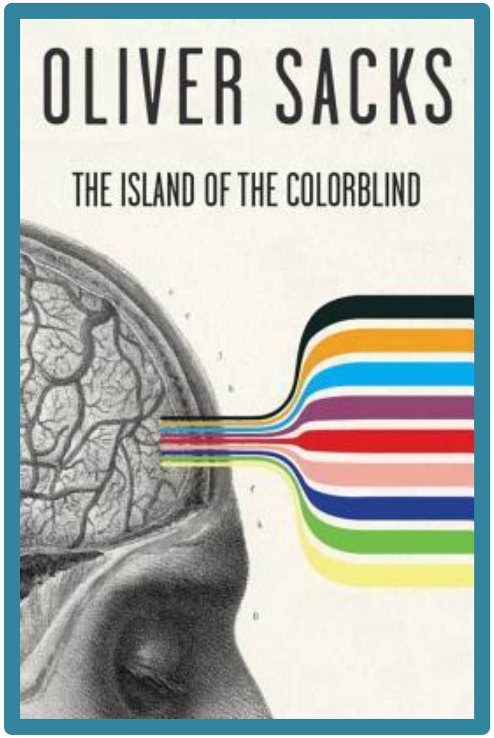
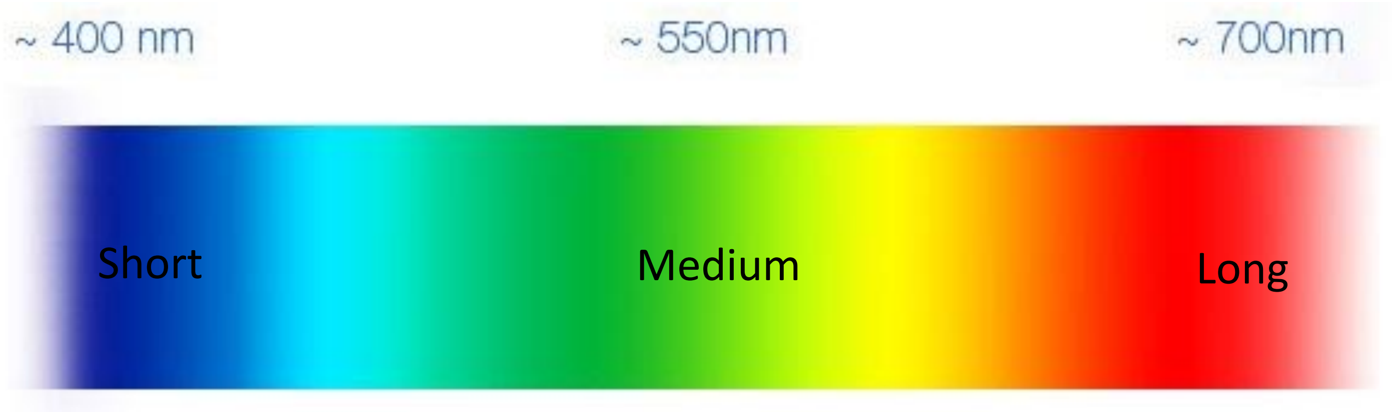


What is colour?

- The visible light spectrum



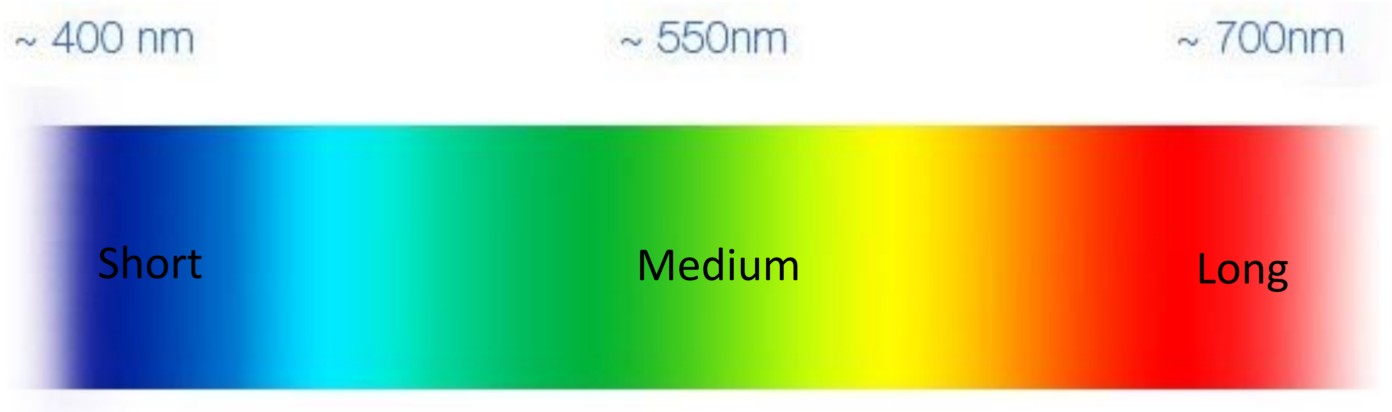
Different wavelengths of the light spectrum are **perceived** as different colours.

Light does not contain colour - colour is non-veridical!

Mausfeld, R. & Heyer, D. (2003). *Colour perception: Mind and the physical world*. Oxford University Press.

What is colour?

- The visible light spectrum



Different wavelengths of the light spectrum are **perceived** as different colours.

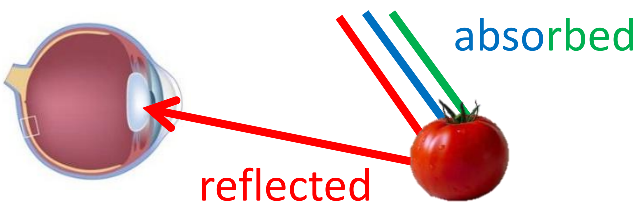
Wavelengths	Perceived colour
Short (& Medium)	Blue
Medium	Green
Medium & Long	Yellow
Long	Red

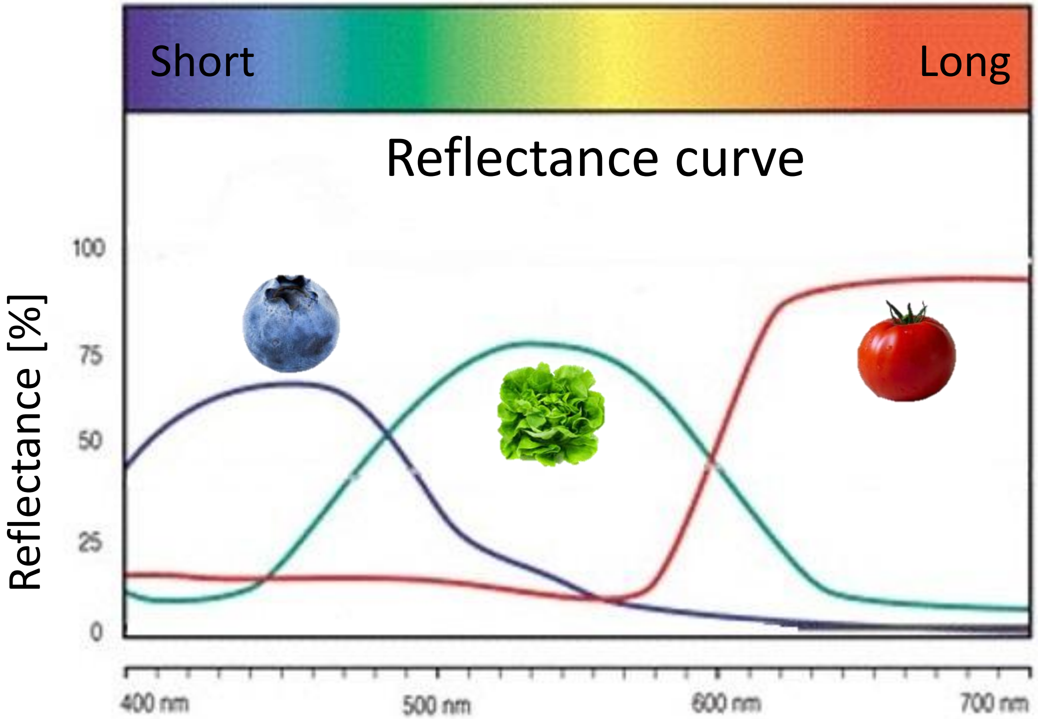
What is colour?

- Colours of **solid objects**
  - Reflected wavelengths

    Chromatic colours are the result of **selective reflection**.

    i.e., Some wavelengths are reflected, others are absorbed.



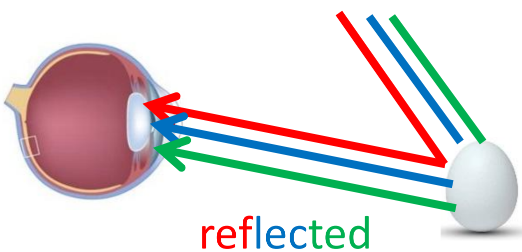


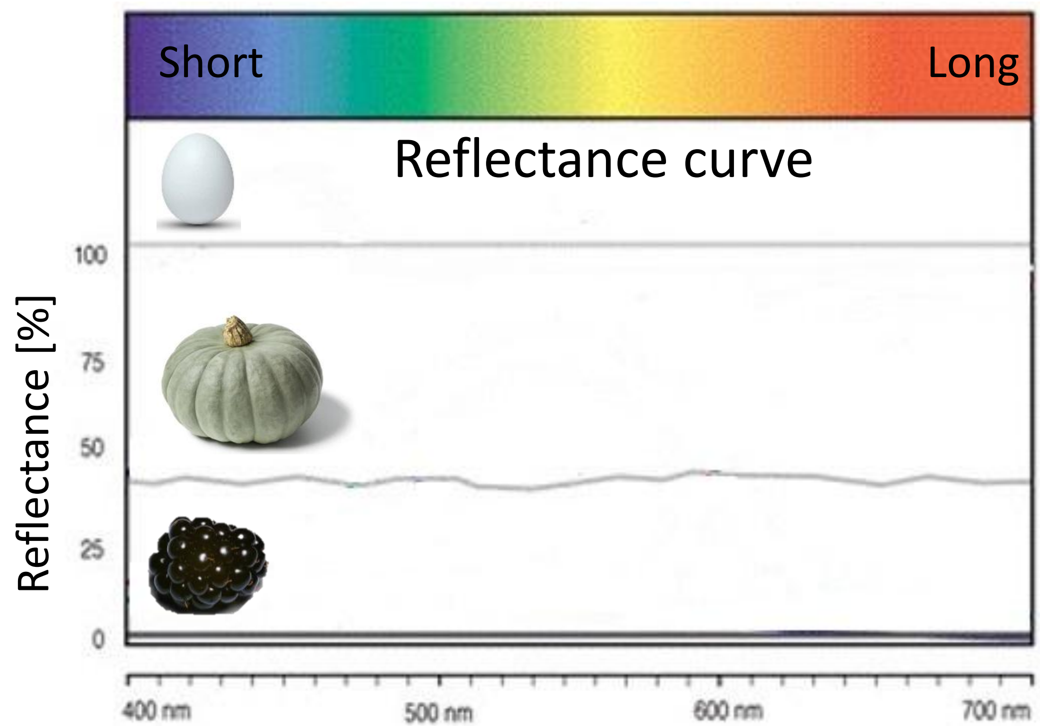
What is colour?

- Colours of **solid objects**
  - Reflected wavelengths

    Achromatic colours are the result of **equal reflection**.

    i.e., All wavelengths are reflected.



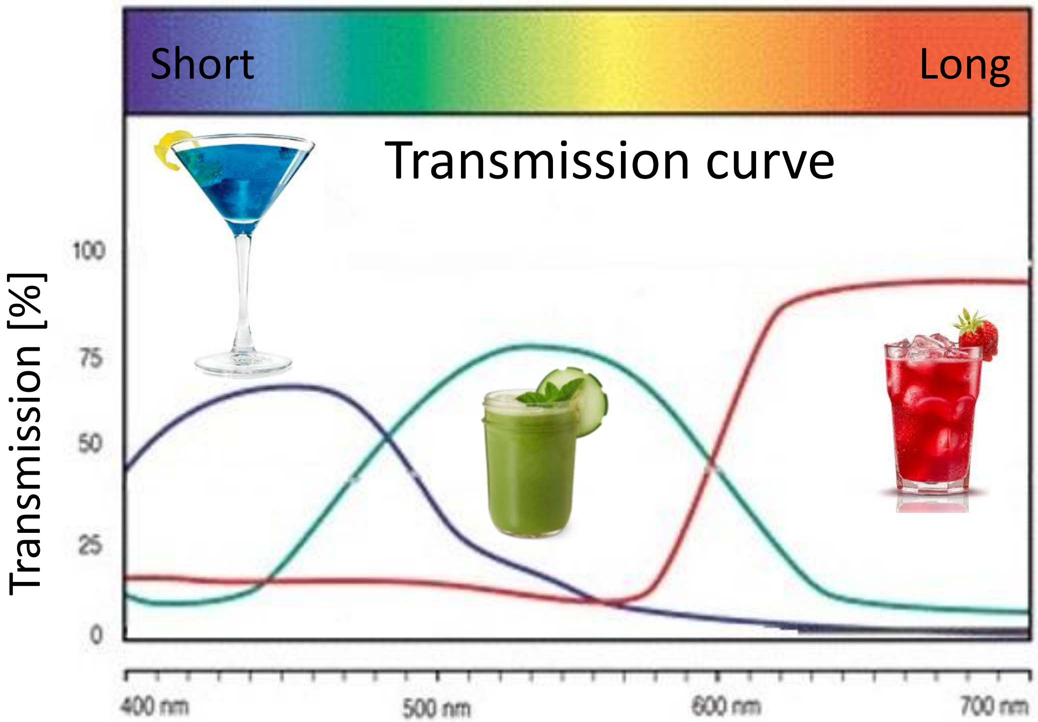


What is colour?

- Colours of **transparent objects**
  - Transmitted wavelengths

    Chromatic colours are the result of **selective transmission**.

    i.e., Some wavelengths are transmitted, others are absorbed.



What is colour?

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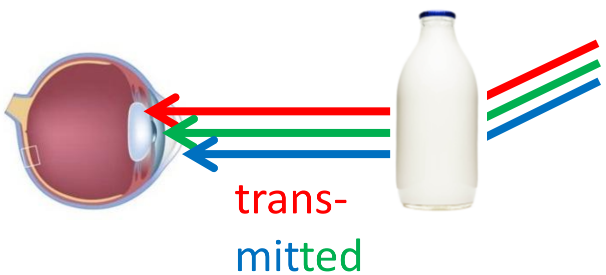
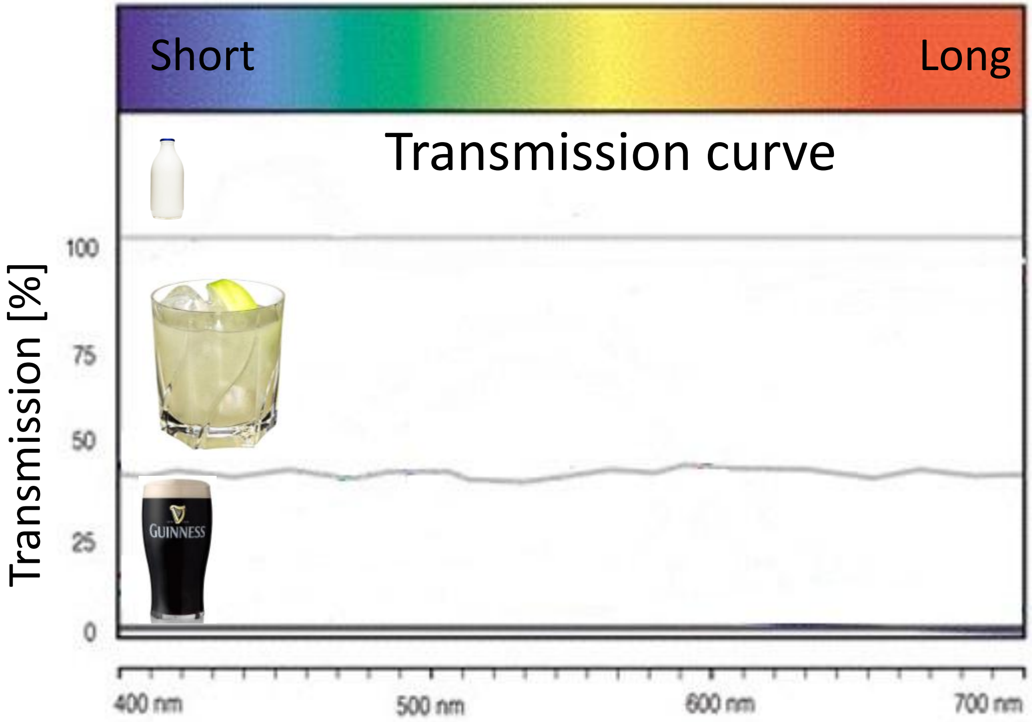


How about water?



How is colour described?

- 3 colour dimensions

  - **Hue** (colour value)

    Represented on the colour wheel.

    Red, green, blue, yellow = pure colours.

    ~200 hues can be discriminated.

  - **Saturation**

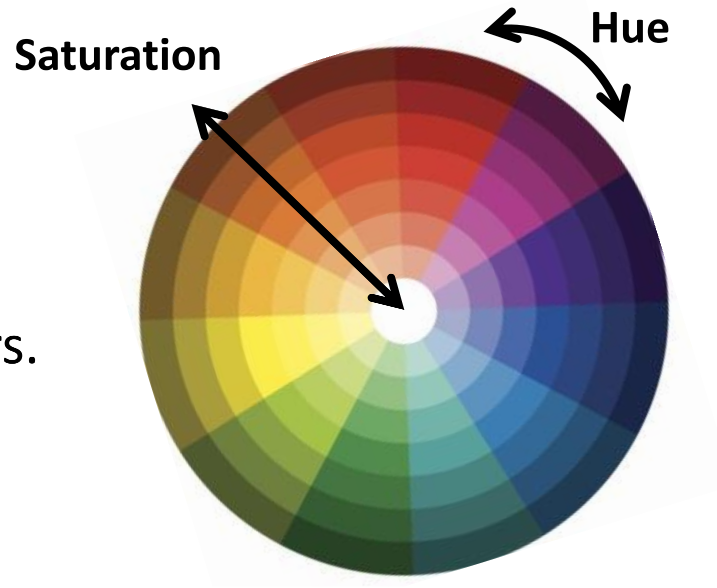
    The amount of white added to the colour.

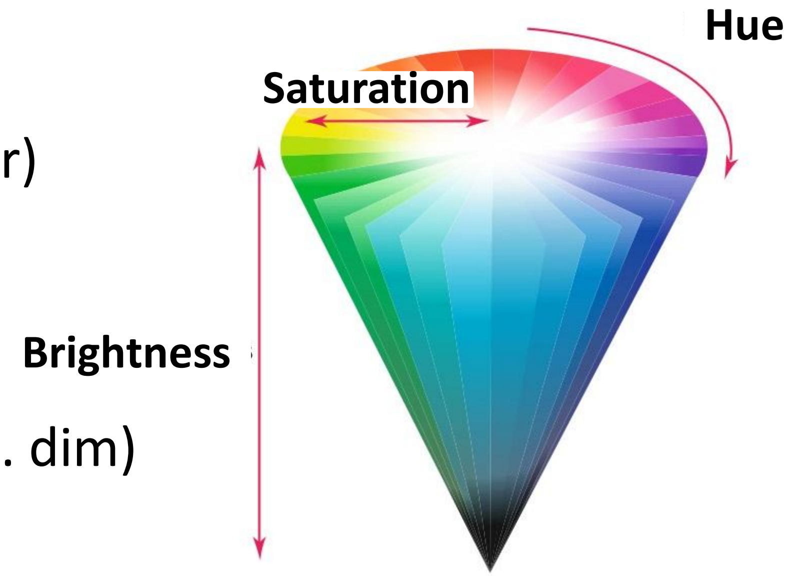
    (saturated vs. non-saturated colour)

  - **Brightness**

    The amount of light reflected.

    The intensity of the light (bright vs. dim)



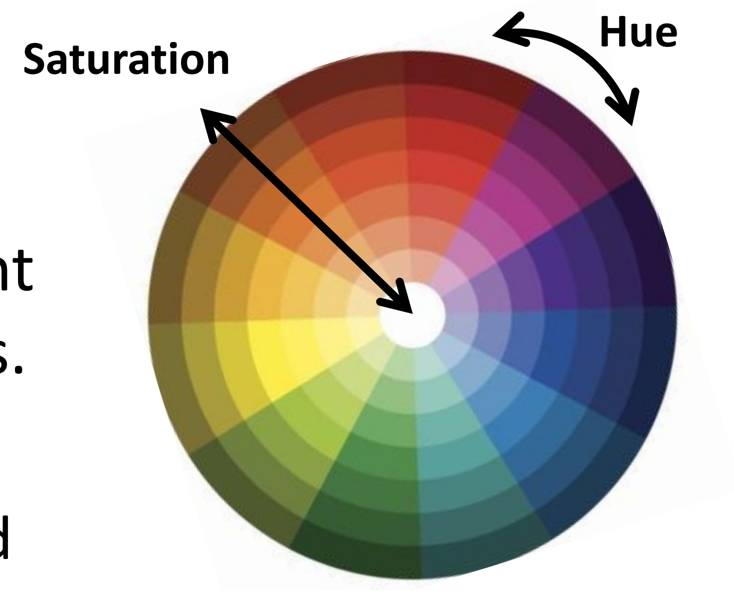


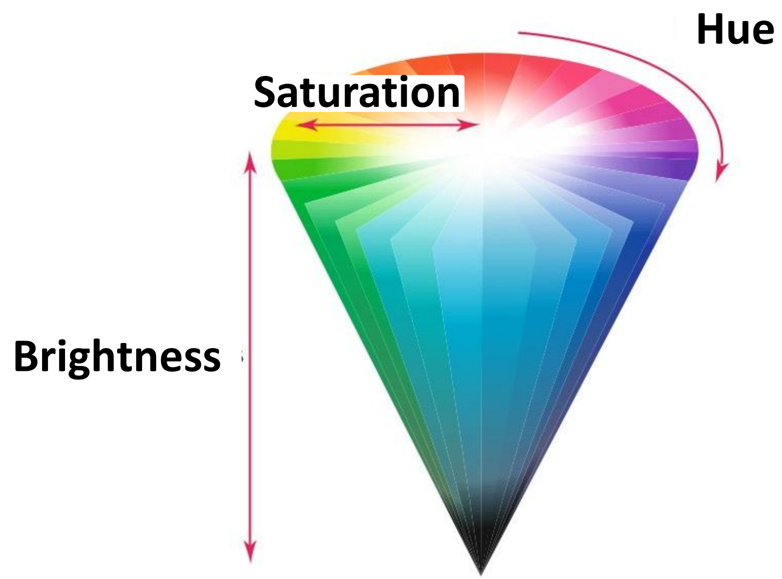
How is colour described?

- 3 colour dimensions
  - We can create **~7 million** different colours along these 3 dimensions.

    The Bureau of Standards identified 7500 colour names (1976).

    http://nvlpubs.nist.gov/nistpubs/Legacy/SP/nbsspecialpublication440.pdf

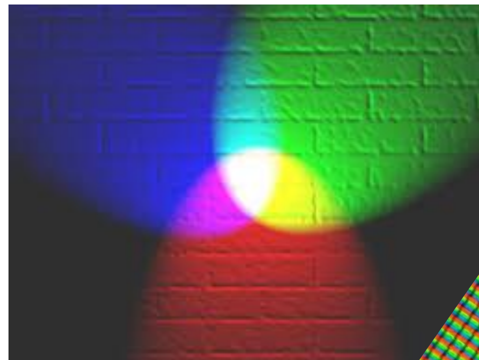


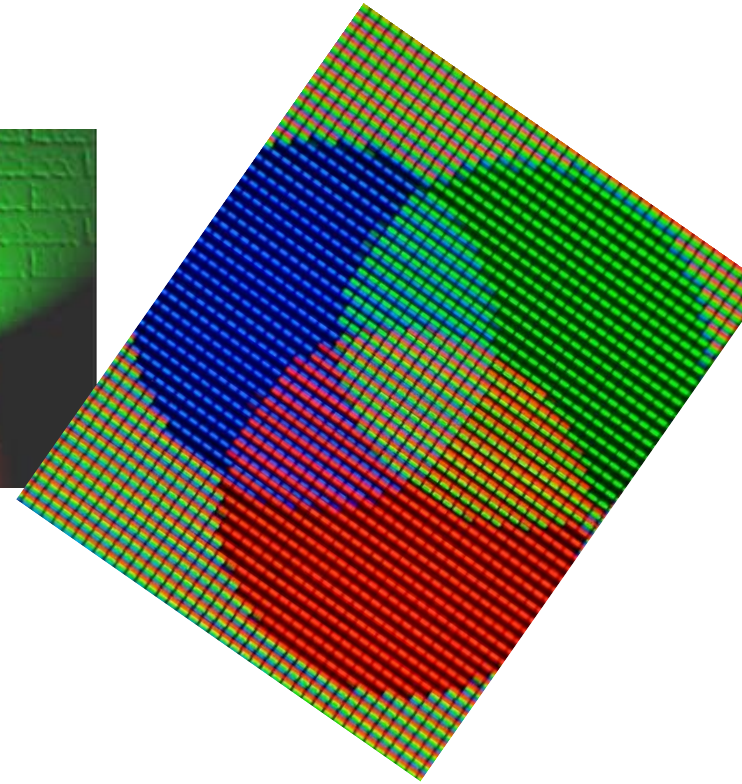


How are colours mixed?

- Mixing light

  e.g., in a display.





- Mixing pigments

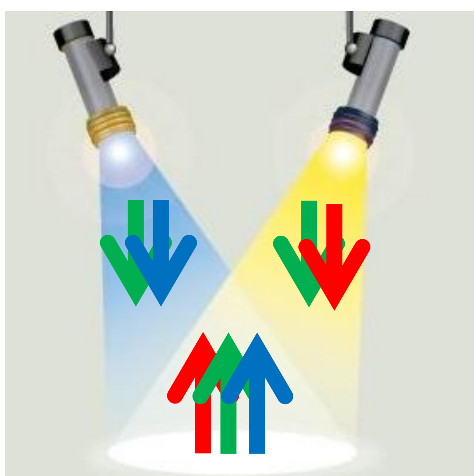
  e.g., in paints.

How are colours mixed?

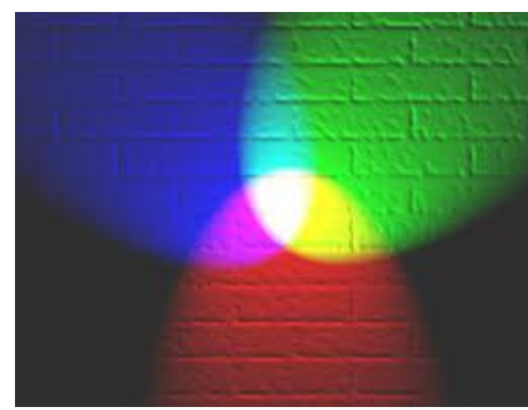
– Mixing **light** = **additive** colour mixture.

All wavelengths present alone are also present when other wavelengths are superimposed.

Addition: There are more wavelengths reflected with two lights as compared to one light.



Short
Medium
Long

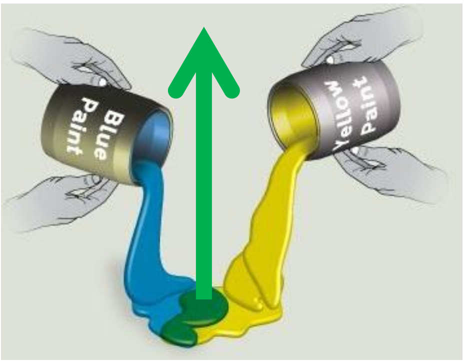


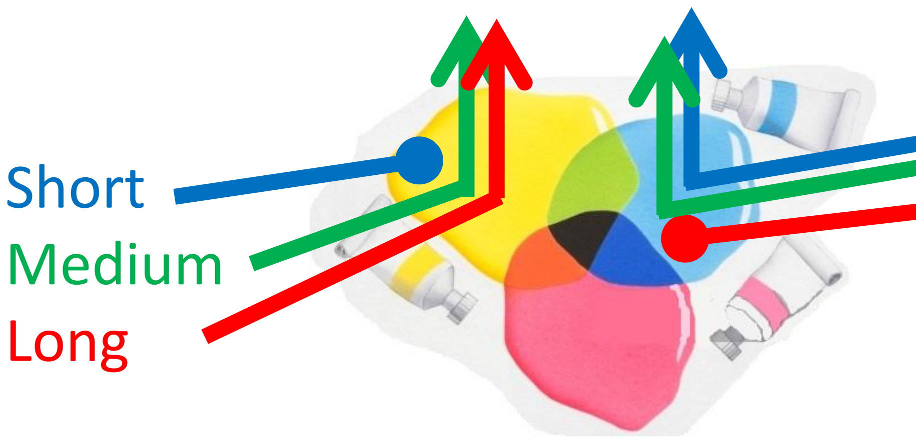
How are colours mixed?

– Mixing **pigments** = **subtractive** colour mixture.

Pigments still absorb the same wavelengths they absorb when they are alone - only the reflected wavelengths remain.

Subtraction: There are less wavelengths reflected with two paints, because together they absorb more wavelengths.



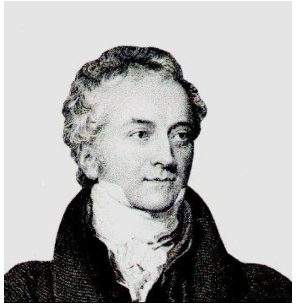


Theories of colour vision

1. Trichromatic theory
2. Opponent process theory

Theories of colour vision (1)

- Trichromatic theory
  - Thomas Young (1802)
    Hermann von Helmholtz (1852)
    (Young-Helmholtz theory of colour vision)



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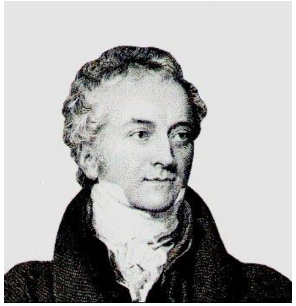
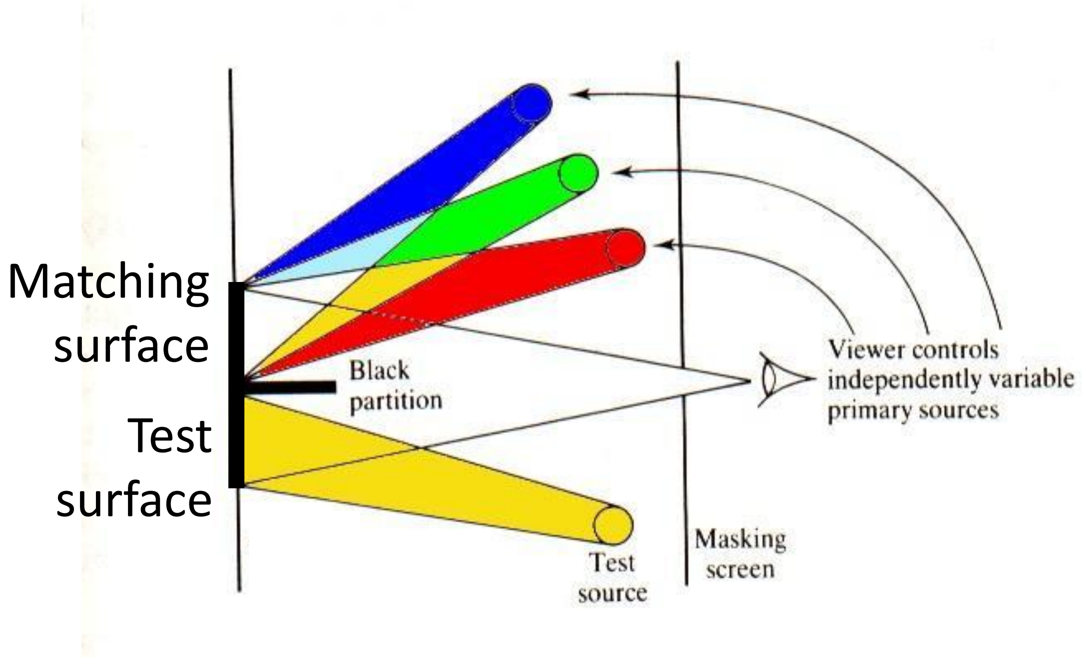


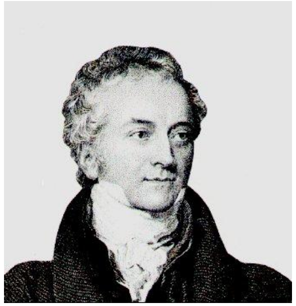
  - **Colour-matching** experiments (Helmholtz)



Task
Combine three lights so that their colour matches the colour shown on the test surface.

Theories of colour vision (1)

- Trichromatic theory
  - Thomas Young (1802)
    Hermann von Helmholtz (1852)



  - **Results**

    1. **Any colour** (in the test field) could be matched by correctly adjusting the proportions of **three different wavelengths**.

    2. Not all colours in the spectrum could be matched with only two different wavelengths.

  - **Conclusion**

    Full colour vision is based on the ability to combine three different wavelengths.

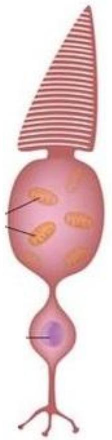
Theories of colour vision (1)

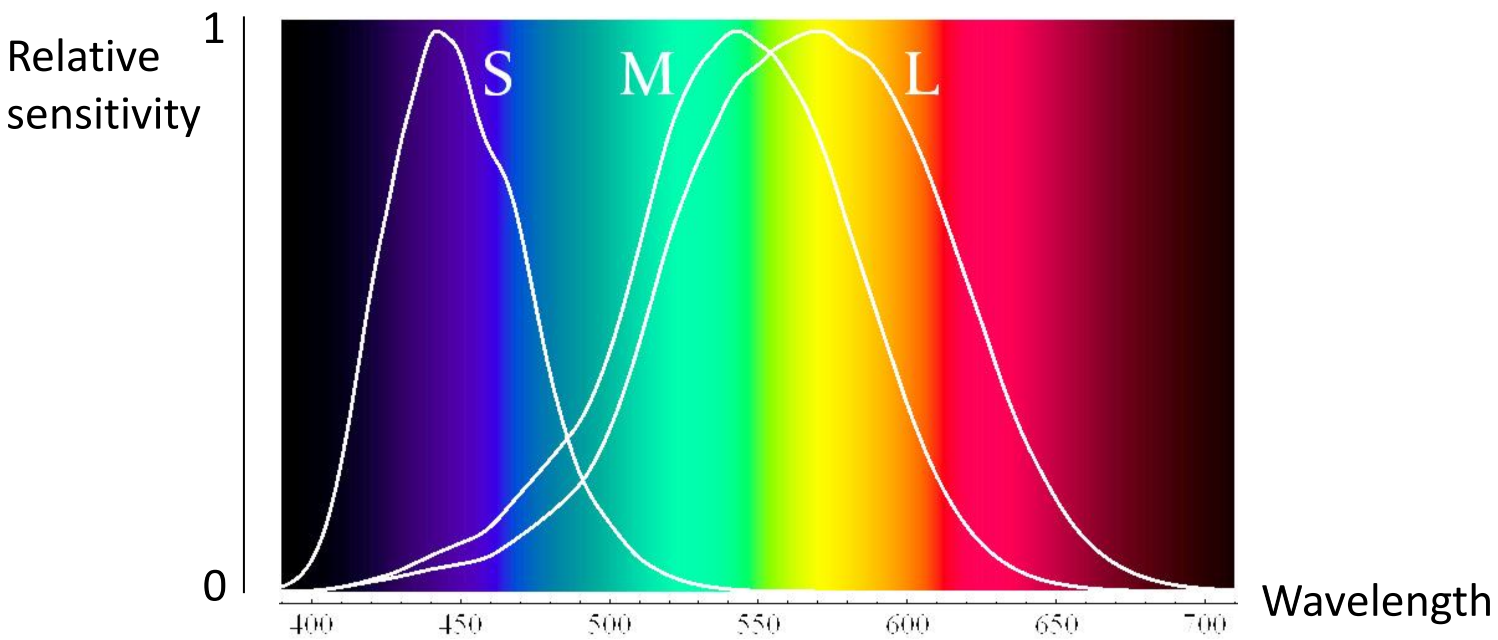
- Trichromatic theory
  - Physiological evidence

    S cones (respond to short wavelengths) blue

    M cones (respond to medium wavelengths) green

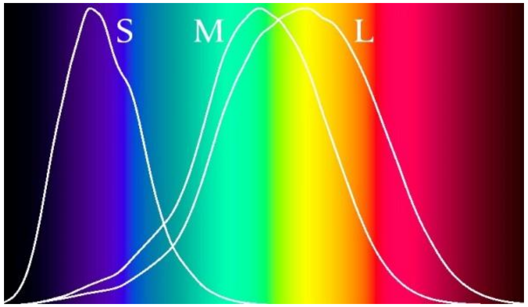
    L cones (respond to long wavelengths) red





Theories of colour vision (1)

- Trichromatic theory
  - Physiological evidence



Wavelength	Perceived colour	Cone types responding		
		S	M	L
Short (& Medium)	Blue	⬆	↑	
Medium	Green	↑	⬆	↑
Medium & Long	Yellow		⬆	⬆
Long	Red		↑	⬆
Long, Medium, & Short	White	⬆	⬆	⬆

Theories of colour vision (1)

- Trichromatic theory
  - Conclusion

    Colour vision is based on the activity of **three different receptor mechanisms** (three different cone types).

    = **Trichromatism**.

  - What happens if people don’t have three different receptor mechanisms?

Impaired colour vision

- Monochromatism

  No cones at all.

- Dichromatism

  Only two types of cones.

To find out how frequent Mono- and Dichromatism is and how it is linked to sex look at: https://nei.nih.gov/health/color_blindness/facts_about

For a colour blind and colour deficiency simulator, as well as detailed information on how to test for colour deficiency, check: http://www.color-blindness.com/

Impaired colour vision

- Monochromatism (Colour blindness)
  - No functioning cones (**rod vision** = scotopic vision).
  - Perceive **shades of brightness** (white, grey, black).

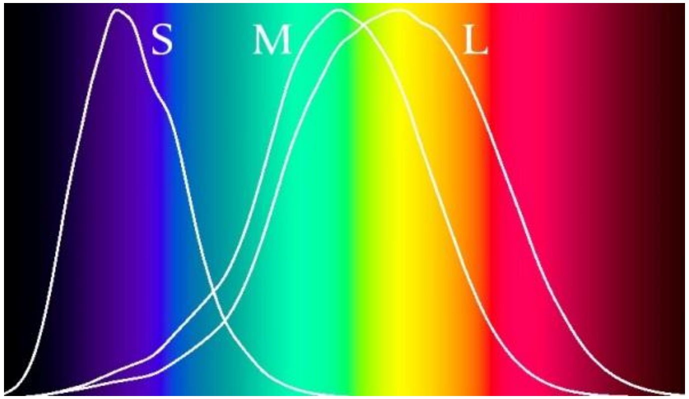
Trichromat

Monochromat

Impaired colour vision

- Dichromatism (Colour deficiency)

Trichromat



  - Protanopia (**no L cones**)

Red/green blindness

  - Deuteranopia (**no M cones**)

  - Tritanopia (**no S cones**)

Blue/yellow blindness

Impaired colour vision

Trichromat

- Dichromatism (Colour deficiency)

  – Protanopia (**no L cones**)

Red/green blindness

  – Deuteranopia (**no M cones**)

  – Tritanopia (**no S cones**)

Blue/yellow blindness

Theories of colour vision (1)

- Trichromatic theory
  - Conclusion

Full Colour vision is based on the activity of **three different receptor mechanisms** (3 different cone types).

= **Trichromatism**.

Support: Impaired colour vision in monochromats and dichromats.

PS

- **Enhanced colour vision?**

- **Tetrachromacy**

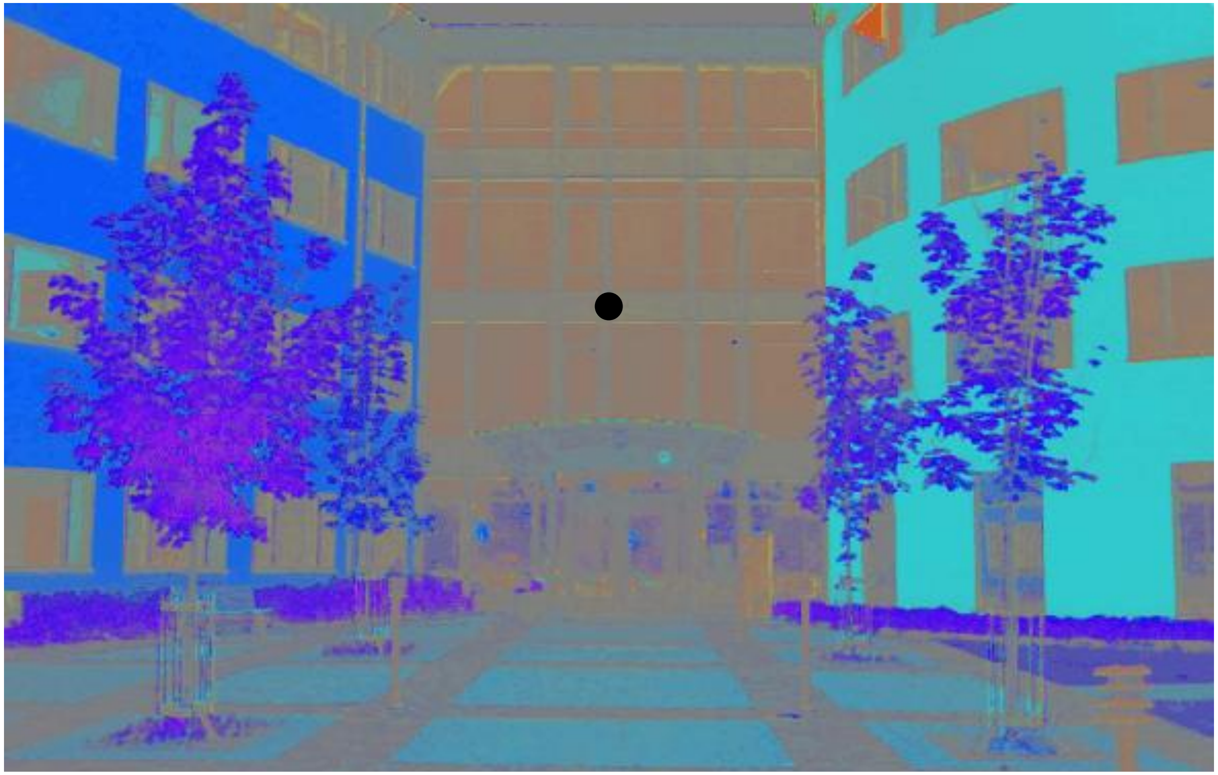
Some animals have **four types of cones**, and thus a four-dimensional colour space.

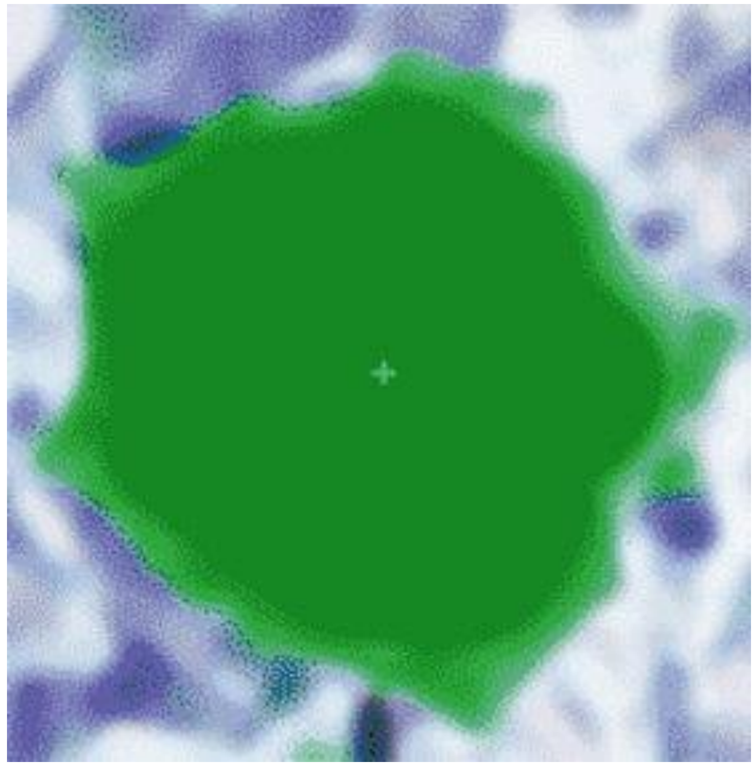
For example, goldfish have four cone types which are sensitive to red, green, blue, and ultraviolet light.

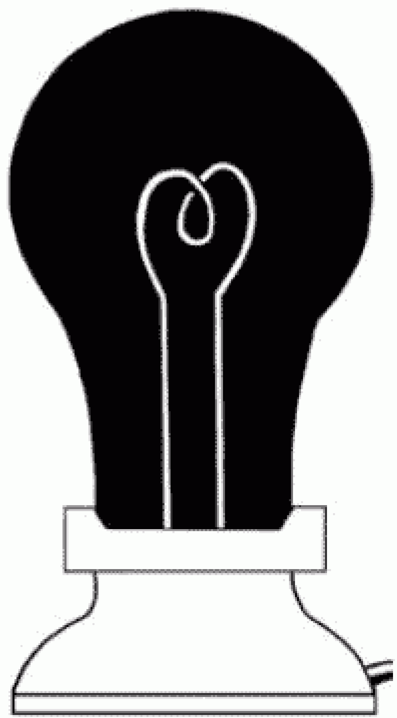
Tetrachromacy has also been described in humans.
See for example this single case study:
https://pdfs.semanticscholar.org/313c/ebb88197fba22a2ec96fdcea9171395268da.pdf

Theories of colour vision (2)

- Opponent-process theory
  - Ewald Hering (1878)







If you like afterimages in general, look at these sites:
http://gpuzzles.com/optical-illusions/after-images/
http://mesosyn.com/mental8-15.html

Theories of colour vision (2)

- Complementary afterimages

  Adapting to **blue** generates a **yellow** afterimage.

  Adapting to **green** generates a **red** afterimage.

  Adapting to **black** generates a **white** afterimage. And vice versa.

  - Furthermore

    Try to imagine a **reddish green**, a **yellowish blue**, a **blackish white**.

  - And Protanopia/Deuteranopia

    There are selective **red-green** or **blue-yellow** colour deficiencies.

    Tritanopia

  - **Conclusion**

    Red & green, blue & yellow, black & white seem to be paired.

Theories of colour vision (2)

- Opponent-process theory
  - Three **opponent channels**

    which respond in opposite ways to different wavelengths.

    Black/white

    i.e., black-on/white-off; black-off/white-on

    Red/green

    i.e., red-on/green-off; red-off/green-on

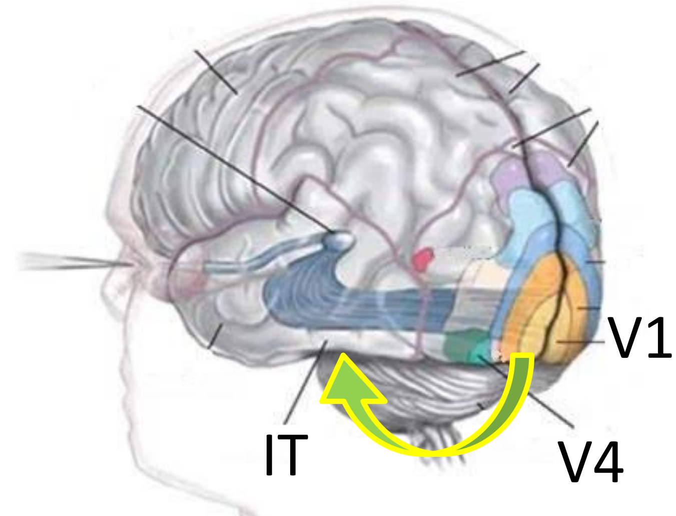
    Blue/yellow

    i.e., blue-on/yellow-off; blue-off/yellow-on

Theories of colour vision (2)

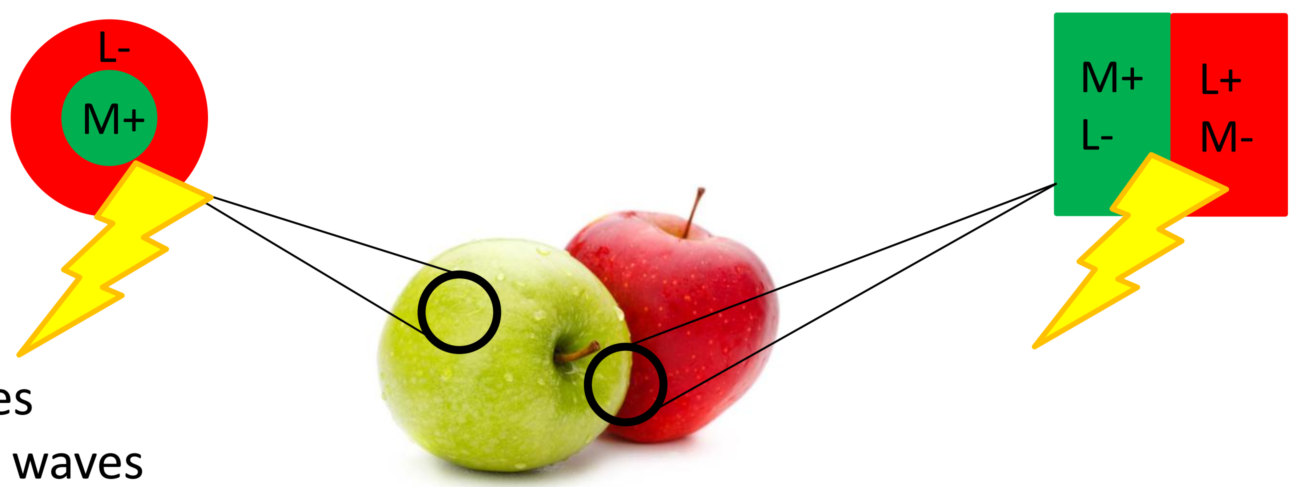
- Opponent-process theory
  - Physiological evidence

    Opponent neurons in V1, V4, and Inferotemporal cortex…



Ventral (what) stream

… with **single-opponent** or **double-opponent** receptive fields.

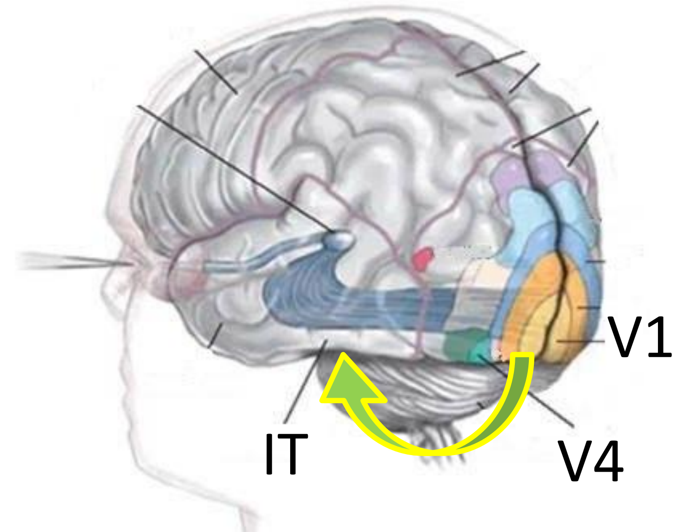


L = long waves
M = medium waves

Theories of colour vision (2)

- Opponent-process theory
  - Physiological evidence

    Opponent neurons in V1, V4, and Inferotemporal cortex…



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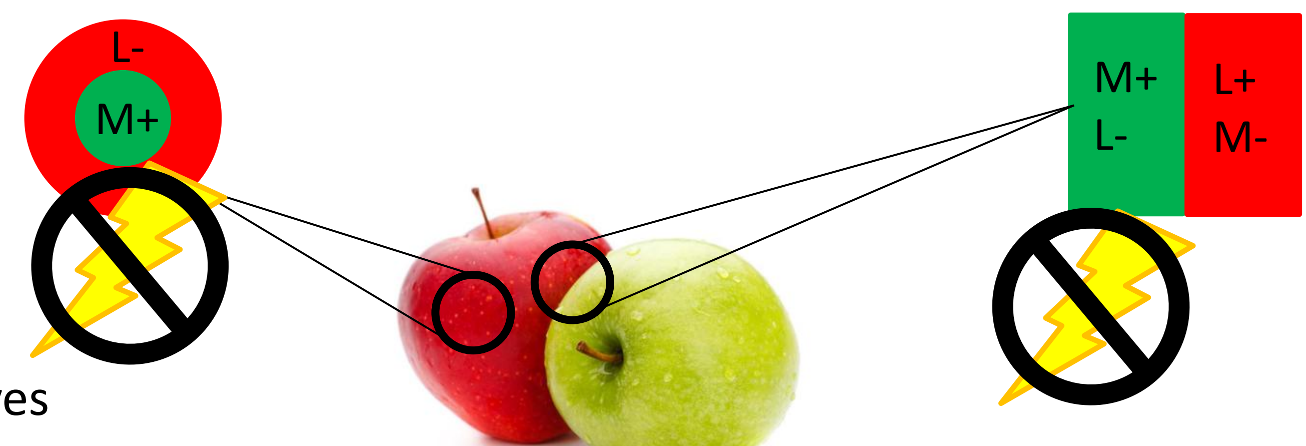


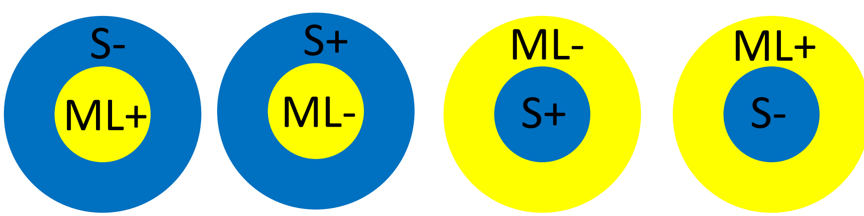
L = long waves
M = medium waves

Theories of colour vision (2)

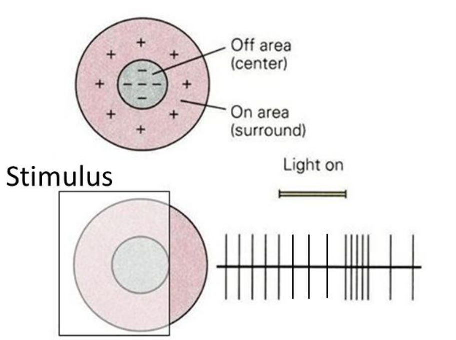
- Opponent-process theory
  - Same for the blue/yellow opponent channel

    Opponent neurons with

    single-opponent receptive fields



ML = medium & long waves
S = short waves
+ = excitation
- = inhibition

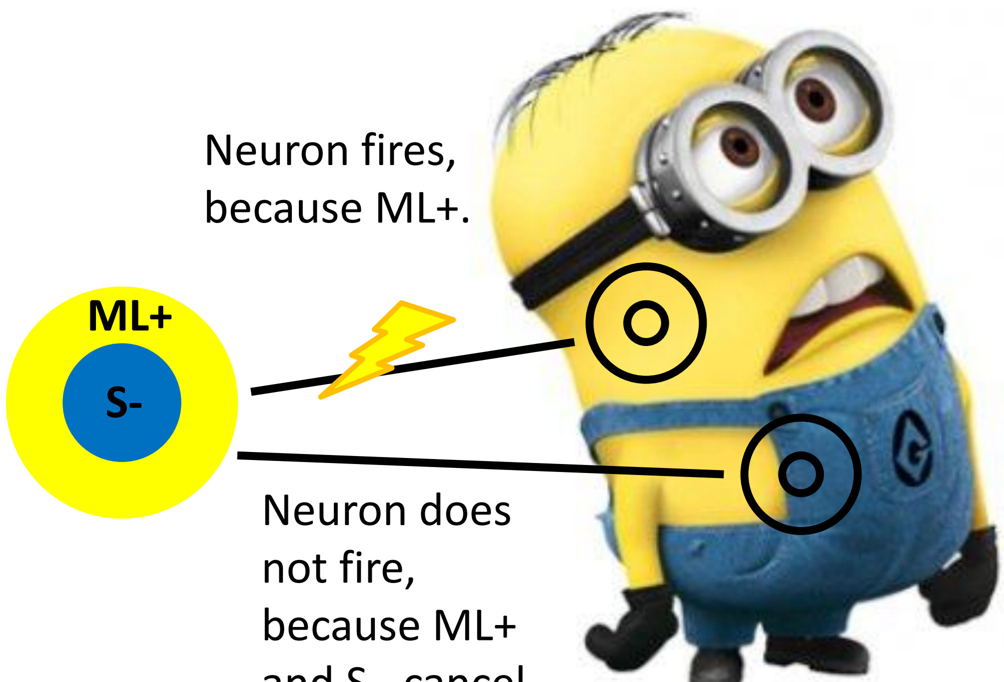


Note: Not to be confused with simple on/off centre/surround receptive fields of ganglion cells which pick up luminance.

Theories of colour vision (2)

- Opponent-process theory

Neurons with **single-opponent** receptive fields are good at **detecting uniform colour surfaces**.

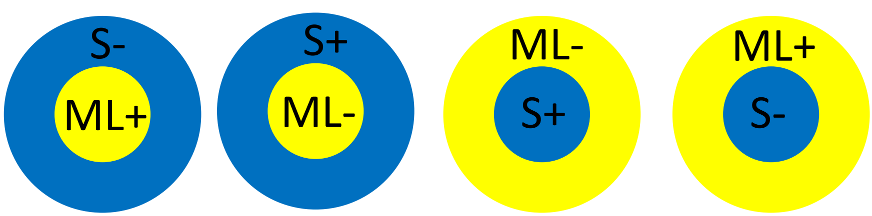


Theories of colour vision (2)

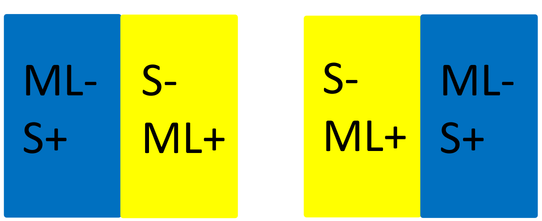
- Opponent-process theory
  - Same for the blue/yellow opponent channel

    Opponent neurons with

    single-opponent receptive fields



double-opponent receptive fields

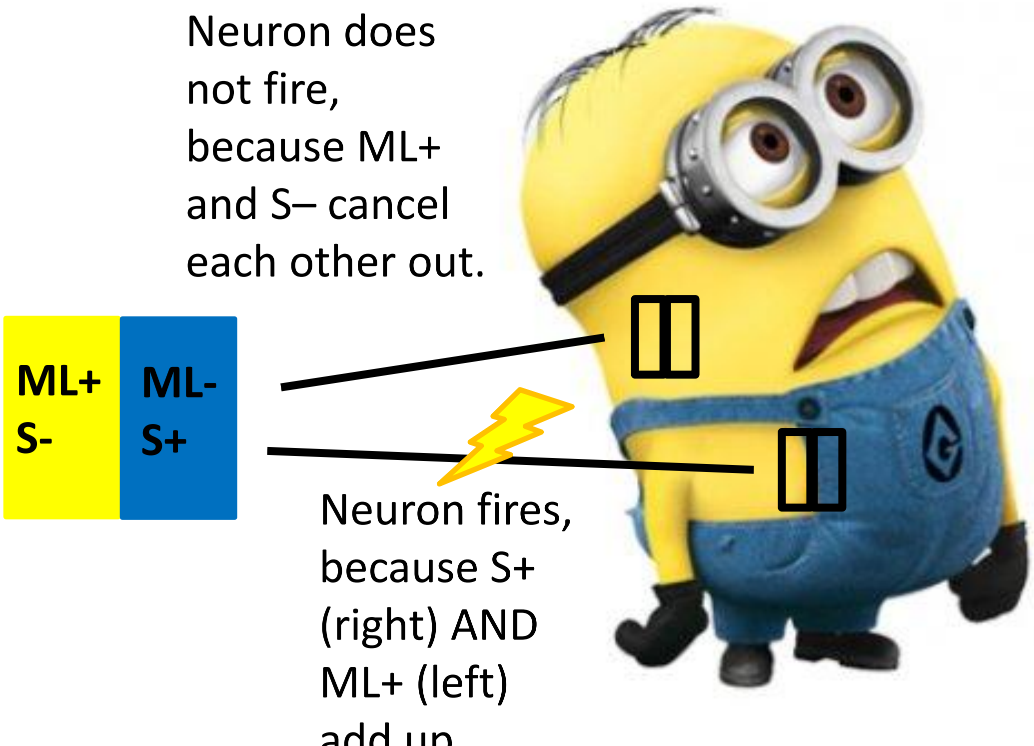


ML = medium & long waves
S = short waves
+ = excitation
- = inhibition

Theories of colour vision (2)

- Opponent-process theory

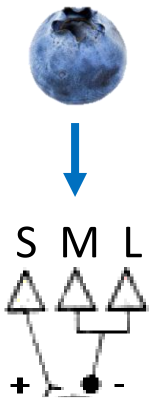
Neurons with **double-opponent** receptive fields are good at **detecting colour boundaries**.



Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
  - **BOTH** are correct!

Cones



Reflects short wavelengths,
absorbs medium and long ones.

Retina: S cones respond, M and L cones don't.

Theories of colour vision – which one is correct?

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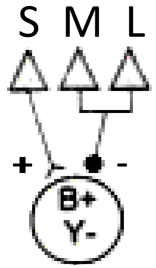
Reflects short wavelengths,
absorbs medium and long ones.

Cones



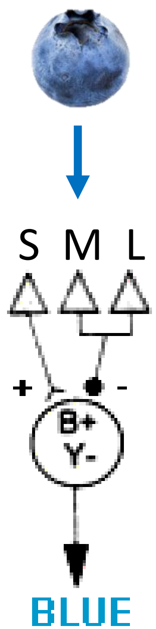
Retina: S cones respond, M and L cones don't.

Opponent
neurons

V1, V4, IT: Blue on/yellow off cells respond.

Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
  - **BOTH** are correct!



Reflects short wavelengths,
absorbs medium and long ones.

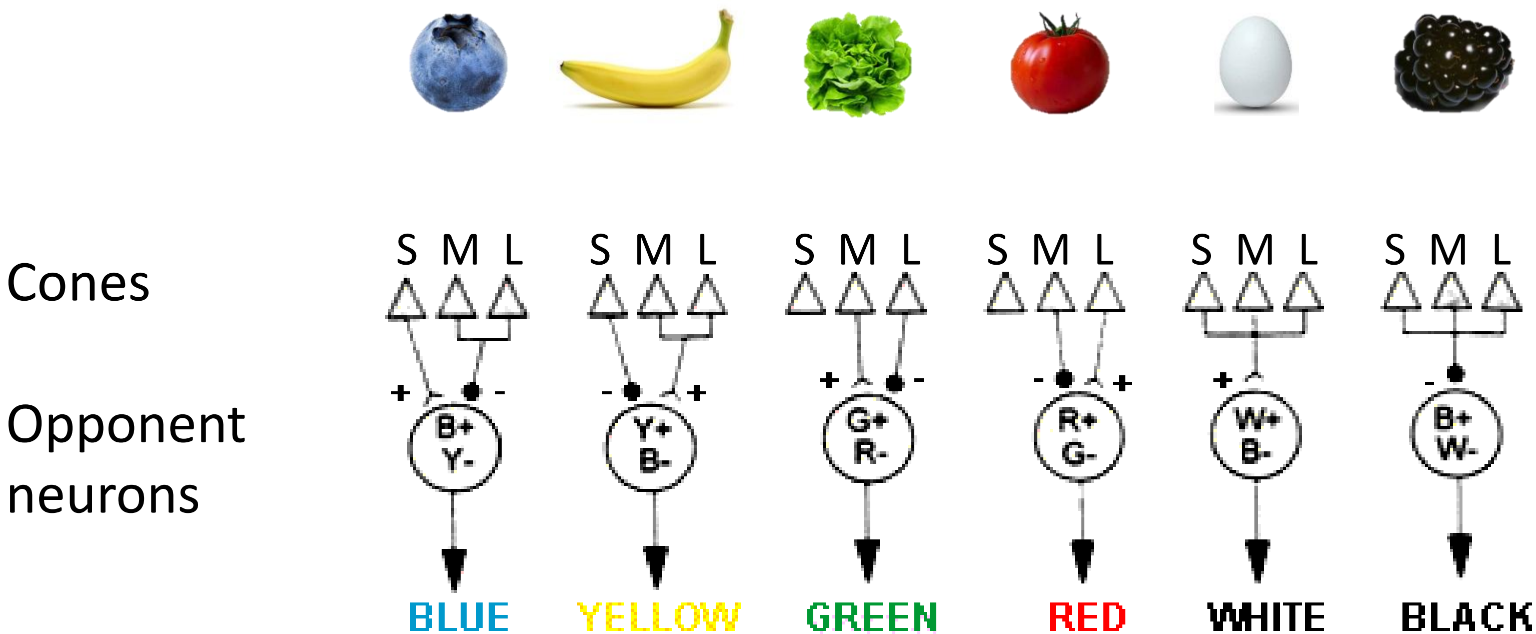
Cones — **Retina**: S cones respond, M and L cones don't.

Opponent neurons — **V1, V4, IT**: Blue on/yellow off cells respond.

Perception

Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
  - **BOTH** are correct!



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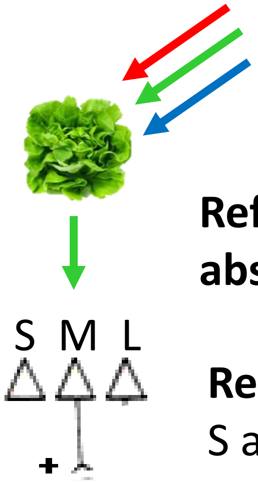
Cones

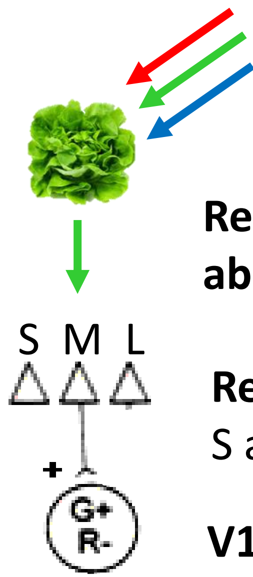


Reflects medium wavelengths, **absorbs** short and long waves.

Retina: M cones respond to medium waves, S and L cones don't.

Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
  - **BOTH** are correct!



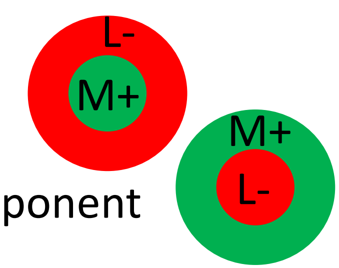
Reflects medium wavelengths,
absorbs short and long waves.

Cones

Retina: M cones respond to medium waves,
S and L cones don't.

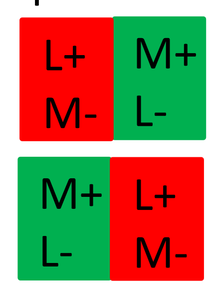
Opponent neurons

V1, V4, IT: Green on/red off cells respond.



Single opponent

Double opponent



Theories of colour vision – which one is correct?

- Trichromatic theory or opponent-process theory?
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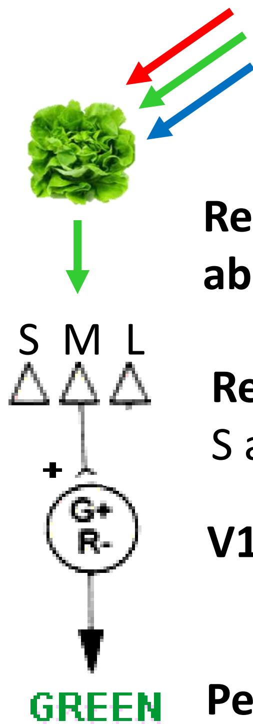

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Cones

Retina: M cones respond to medium waves,
S and L cones don't.

Opponent
neurons

V1, V4, IT: Green on/red off cells respond.

Perception

Summary

- We have looked at
  - Different wavelengths and that they are perceived as different colours.

    Solid objects reflect / transparent objects transmit different wavelengths selectively (chromatic colours) or completely (achromatic colours).
  - Three dimensions (hue, saturation, brightness) to describe different colours.
  - Colour mixtures with light (additive) versus pigments (subtractive).
  - Two theories of colour perception and how they complement each other.

    First step of colour perception is explained by the trichromatic theory (S, M, and L cones are needed for a full colour experience).

    Second step of colour perception is explained by the opponent process theory (opponent neurons in the ventral stream respond selectively to red/green, blue/yellow, and black/white).

Test yourself – learning goals

- After reviewing this lecture, you should be able to
  - Explain what the reflectance and transmission curves show.
  - Explain what is meant by colour hue, saturation, and brightness.
  - Discuss the difference between mixing lights and mixing pigments.
  - Describe the trichromatic and the opponent process theory of colour vision, discuss their physiological evidence, and explain how they complement each other.
  - Explain what a monochromat, dichromat and trichromat is.

Test yourself – multiple choice

1. Why do we perceive a lime as green?
   a) A lime absorbs medium and long wavelengths, but selectively transmits short wavelengths, therefore we see green.
   b) A lime reflects short and long wavelengths, but selectively absorbs medium wavelengths, therefore we see green.
   c) A lime absorbs short and long wavelengths, but selectively reflects medium wavelengths, therefore we see green.
   d) A lime absorbs short and medium wavelengths, but selectively transmits long wavelengths, therefore we see green.

Test yourself – multiple choice

2. From the assumptions of the trichromatic theory of colour vision (Young and Helmholtz) it can be derived that:

   a) Any colour can be matched by any two other wavelengths (in correct proportions).

   b) There is no full colour experience if one or more types of cones are not functioning properly.

   c) There are no colours which can be described as blueish-yellow and reddish-green.

   d) Trichromatism is a matter of repeated colour subtraction (until the correct proportion is found).

Test yourself – multiple choice

3. Which statement about colour vision is wrong?

    a) The opponent stage involves three channels: red-green, blue-yellow, black-white.

    b) The trichromatic stage involves three classes of cones, which are sensitive to short, medium and long wavelengths.

    c) An initial stage of opponent processing is followed by trichromatic analysis.

    d) Yellow-on/blue-off receptive fields of opponent neurons combine excitatory activation from M and L cones.

See you on Friday…